

Two-dimensional conformal field theory

Primary fields, the stress tensor, and finite-size energies

Conformal symmetry constrains correlations without requiring a microscopic solution. This note concentrates on a local two-dimensional Euclidean CFT on the plane, with complex coordinates $z = x + iy$ and $\bar{z} = x - iy$. It is a small set of tools for reading calculations, rather than a classification of CFTs.

1 Fields with definite scaling weights

A locally conformal coordinate change has $w = f(z)$ and $\bar{w} = \bar{f}(\bar{z})$. A primary field ϕ of weights $(h, \bar{h})$ transforms in the passive convention as

$$\phi'(w, \bar{w}) = \left(\frac{dz}{dw}\right)^h \left(\frac{d\bar{z}}{d\bar{w}}\right)^{\bar{h}} \phi(z, \bar{z}). \quad (1)$$

Its scaling dimension is $\Delta = h + \bar{h}$ and its spin is $s = h - \bar{h}$. Taking $w = \lambda z$ gives the factor $\lambda^{-\Delta}$ for real positive λ .

For a normalized primary and its conjugate with the same weights, translation, rotation, scaling, and special conformal symmetry fix the plane two-point function to

$$\langle \phi(z, \bar{z}) \phi^\dagger(0) \rangle = \frac{1}{z^{2h} \bar{z}^{2\bar{h}}}. \quad (2)$$

Two-point functions between distinct conformal representations vanish in a diagonal basis. Three-point functions are similarly fixed up to constants. Four-point functions retain nontrivial dependence on conformal cross ratios, so symmetry alone does not determine all dynamics.

2 The stress tensor generates transformations

Write $T(z)$ for the holomorphic stress tensor and c for its central charge. The singular terms in its operator product expansions are

$$T(z)\phi(w, \bar{w}) \sim \frac{h\phi(w, \bar{w})}{(z-w)^2} + \frac{\partial_w \phi(w, \bar{w})}{z-w}, \quad (3)$$

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w}. \quad (4)$$

The symbol $\sim$ here displays singular terms as $z \rightarrow w$. Integrating T around an insertion gives the conformal Ward identity. Expanding $T(z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$ and applying nested contours to Eq. (4) yields the Virasoro algebra

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}. \quad (5)$$

There is an independent antiholomorphic copy with generators $\bar{L}_n$ and central charge $\bar{c}$. The two copies commute.

3 States, descendants, and one example

In radial quantization, the logarithm of radius acts as Euclidean time. A local primary at the origin creates a state satisfying

$$L_0|h, \bar{h}\rangle = h|h, \bar{h}\rangle, \quad L_n|h, \bar{h}\rangle = 0 \quad (n > 0), \quad (6)$$

with analogous barred equations. Negative modes create descendants; L_{-n} increases the holomorphic weight by n . This packages infinitely many local operators into representations.

For a normalized primary state in a unitary theory, $L_n^\dagger = L_{-n}$. Equation (5) immediately gives

$$\|L_{-1}|h, \bar{h}\rangle\|^2 = \langle h, \bar{h}|[L_1, L_{-1}]|h, \bar{h}\rangle = 2h. \quad (7)$$

Thus $h \geq 0$; the barred sector gives $\bar{h} \geq 0$. For the vacuum, translation invariance makes the level-one descendant null.

Critical Ising example. The local spin field has $(h, \bar{h}) = (1/16, 1/16)$ in a theory with $c = \bar{c} = 1/2$. Therefore $\Delta = 1/8$ and its normalized two-point function decays as $|z|^{-1/4}$. Confusing h with Δ would give the wrong critical exponent.

4 From the plane to a cylinder

Let $w = v\tau + ix$ with $x \sim x + L$, where L is the spatial circumference and v the propagation velocity. The exponential map is $z = \exp(2\pi w/L)$. The stress tensor transforms with an additional Schwarzian term:

$$T_{\text{cyl}}(w) = \left(\frac{dz}{dw}\right)^2 T_{\text{plane}}(z) + \frac{c}{12}\{z, w\}, \quad (8)$$

$$\{z, w\} = \frac{z'''}{z'} - \frac{3}{2}\left(\frac{z''}{z'}\right)^2 = -\frac{1}{2}\left(\frac{2\pi}{L}\right)^2. \quad (9)$$

The plane vacuum has $\langle T \rangle = 0$, but the cylinder vacuum has a nonzero stress. Restoring $\hbar$, the CFT Hamiltonian is

$$H = \frac{2\pi\hbar v}{L} \left(L_0 + \bar{L}_0 - \frac{c + \bar{c}}{24} \right). \quad (10)$$

For $c = \bar{c}$, the vacuum contribution is $E_0 = -\pi\hbar v c/(6L)$. A microscopic periodic critical chain also has a nonuniversal extensive ground-state energy; this formula gives its universal $1/L$ correction when the vacuum sector is selected. Excitation gaps determine scaling dimensions through $E - E_0 = 2\pi\hbar v \Delta/L$.

References

- [1] P. Ginsparg, *Applied Conformal Field Theory*, Les Houches lectures (1988). [arXiv:hep-th/9108028](#).
- [2] J. D. Qualls, *Lectures on Conformal Field Theory* (2015). [arXiv:1511.04074](#).